

WaterTAP Costing Reference Guide

Detailed Unit Model Costing Parameters

These models compute costs from physics-derived sizing variables. Capital cost equations vary by unit type. All parameters are tunable.

Unit Model	Capital Cost Eq.	Key Default Parameters	Reference
Anaerobic Digester	$A \times (Q/Q_{ref})^B$	A = \$19.36M B = 0.6 $Q_{ref} = 911 \text{ m}^3/\text{day}$	NREL Waste-to-Energy Model
CSTR	$A \times V^B$	A = \$1,246.1/m ³ B = 0.71	C.C. Tang, 1984
Chiller	$C_{chill} \times (W_{duty}/COP)$	$C_{chill} = \$200/\text{kW}$ COP = 7	
Clarifier (Circular)	$A \times A_{surf}^2 + B \times A_{surf} + C$	A = -6×10^{-4} B = 98.952 C = 191,806	Sharma et al. 2013
Clarifier (Primary)	$A \times Q_{MGD}^B$	A = \$538,746 B = 0.7	Byun et al. 2022
Clarifier (Rectangular)	$A \times A_{surf}^2 + B \times A_{surf} + C$	A = -2.9×10^{-3} B = 169.19 C = \$94,365	Sharma et al. 2013
Compressor	$C_{comp} \times W_{mech}^B$	$C_{comp} = \$7364$ B = 0.7	El-Sayed et al., 2001
Crystallizer	$A \times (Q/Q_{ref})^B \times IEC$	A = \$675,000 $Q_{ref} = 1 \text{ m}^3/\text{hr}$ B = 0.53 IEC = 1.43 steam = \$0.004/kg	Woods, 2007; Diab and Gerogiorgis, 2017; Yusuf et al., 2019; Panagopoulos (2019)
Dewatering (Centrifuge)	$A \times Q + B$	A = \$328.03/(gal/hr) B = \$751,295	McGivney & Kawamura, 2008
Dewatering (Filter Belt Press)	$A \times Q + B$	A = \$146.29/(gal/hr) B = \$433,972	McGivney & Kawamura, 2008
Dewatering (Filter Plate Press)	$A \times Q^B$	A = \$102,784/(gal/hr) B = 0.4216	McGivney & Kawamura, 2008
Electrodialysis	$C_{mem} \times A_{mem} + C_{electrode}$	$C_{mem} = \$160/\text{m}^2$ $C_{electrode} = \$2100/\text{m}^2$ replacement = 20 %/yr (mem + electrode)	
Electrolyzer	$C_{mem} + C_{anode} + C_{cathode}$	$C_{mem} = \$25/\text{m}^2$ $C_{anode} = \$300/\text{m}^2$ $C_{cathode} = \$600/\text{m}^2$ mat fraction = 65 %	Desalination 452 (2019) 265–278
Energy Recovery Device	$C_{ERD} \times Q$	$C_{ERD} = \$535/(\text{m}^3/\text{s})$	
Evaporator	$C_{evap} \times A_{evap}$	$C_{evap} = \$1000/\text{m}^2$ material factor = 1.0	
GAC	$C_{contactor} + C_{gac} + C_{other}$	$C_{contactor} = \text{polynomial } f(\text{vol})$ $C_{gac} = \text{exponential } f(\text{mass})$ $C_{other} = \text{power } f(\text{vol})$ $C_{regen} = \$4.28/\text{kg}$ $C_{makeup} = \$4.58/\text{kg}$	EPA-WBS 2021
Heat Exchanger	$C_{HX} \times A_{hx}$	$C_{HX} = \$300/\text{m}^2$ steam = \$0.008/kg	

Heater (Electric)	$C_{\text{heat}} \times W_{\text{duty}} / \text{eff}$	$C_{\text{heat}} = \$66/\text{kW}$ $\text{eff} = 0.99$	
Ion Exchange (Cation / Anion)	$C_{\text{resin}} + C_{\text{vessels}} + C_{\text{tanks}}$	$C_{\text{res,AX}} = \$205/\text{ft}^3$ $C_{\text{res,CX}} = \$153/\text{ft}^3$ vessel A=1596.5, b=0.46 resin replacement = 5 %/yr	EPA-WBS 2021
Membrane Distillation	$C_{\text{mem}} \times A_{\text{mem}}$	$C_{\text{mem}} = \$56/\text{m}^2$ replacement = 20 %/yr	
Mixer	$C_{\text{mix}} \times Q$	Generic = $\$361/(\text{L/s})$ NaOCl mixer = $\$5.08/(\text{m}^3/\text{day})$ $\text{CaOH}_2 = 873.9/(\text{kg}/\text{day})$	
Nanofiltration	$C_{\text{mem}} \times A_{\text{mem}}$	$C_{\text{mem}} = \$15/\text{m}^2$ replacement = 20 %/yr	
OARO	$C_{\text{mem}} \times A_{\text{mem}}$	$C_{\text{mem}} = \$30/\text{m}^2$ $C_{\text{mem,HP}} = \$50/\text{m}^2$ replacement = 15 %/yr	Bartholomew et al. 2017
Pressure Exchanger	$C_{\text{PX}} \times Q$	$C_{\text{PX}} = \$535/(\text{m}^3/\text{s})$	
Pump (high pressure)	$C_{\text{pump}} \times W_{\text{mech}}$	$C_{\text{pump}} = \$53/\text{W}$	Malek et al. 1996
Pump (low pressure)	$C_{\text{pump}} \times Q$	$C_{\text{pump}} = \$889/(\text{m}^3/\text{s})$	
Reverse Osmosis	$C_{\text{mem}} \times A_{\text{mem}}$	$C_{\text{mem}} = \$30/\text{m}^2$ $C_{\text{mem,HP}} = \$75/\text{m}^2$ replacement = 20 %/yr	Bartholomew et al. 2018
Steam Ejector	$A \times (M)^B$	M = motive steam + entrained vapor; kg/hr A = \$1949 B = 0.3 steam = \$0.008/kg	Gabriel 2015, Desalination
Thickener	$A \times A_{\text{surf}} + B$	A = 4729.8/ft ² B = \$37,068	McGivney & Kawamura, 2008
UV+AOP	$C_{\text{reactor}} + C_{\text{lamp}}$	$C_{\text{reactor}} = \$202.35/\text{kW}$ $C_{\text{lamp}} = \$235.50/\text{kW}$ lamp replacement = 33.3 %/yr	

WaterTAP Costing Reference Guide (continued)

Zero-Order Unit Model Costing — Default Parameters

Zero-order models use: $C_{\text{cap}} = A \times (Q_{\text{in}}/Q_{\text{basis}})^B$. A is in USD for the reference year shown. Costs are CPI-adjusted to the study year.

Unit Model	A	B	Q_{basis}	Cost Year	Energy (kWh/m ³)	Recovery	Reference
Aeration Basin	725,570	0.586	1 MGD	2014	0.412	N/A	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Anaerobic Digestion Oxidation	19,355,231	0.600	911 m ³ /hr	2012	0.15	0.9999	NREL Waste-to-Energy Model (WESyS)
Anaerobic Digestion Reactive	19,355,231	0.600	911 m ³ /hr	2012	0.2	N/A	Unknown
Backwash Solids Handling	41,812	0.918	1 MGD	2007	f(Q)	0.95	Based on costs for subprocesses in Table 5.7.1 (Equations 29, 30, 31, 32, 33, 37, 38, 39) in Cost Estimating Manual for Water Treatment Facilities (McGivney/Kawamura)
Bio-Active Filtration	725,570	0.586	1 MGD	2014	None	0.99	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Bioreactor	1,252,160	1	45 m ³ /hr	2018	None	0.97	Developed with Aspen Cost Estimation for San Luis Case Study in WT3.
Blending Reservoir	2,000,000	0.700	50,000 m ³ /hr	2020	None	N/A	Unknown
Buffer Tank	1,000,000	0.700	29,526 m ³ /hr	2019	None	N/A	Unknown
CO ₂ Addition	464,000	0.700	1 MGD	2019	None	N/A	Unknown
Cartridge Filtration	725,570	0.586	1 MGD	2014	0.0002	0.9999	Texas Water Development Board IT3P Figure 3.3
Clarifier	177,516	0.318	5,450 m ³ /hr	2007	None	0.999	McGivney & Kawamura, Figure 5.5.24c + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Conventional Activated Sludge	725,570	0.586	1 MGD	2014	0.14	0.9999	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Cooling Supply	40,299	0.866	1 MGD	2007	None	N/A	McGivney & Kawamura Figure 5.5.35b
Cooling Tower	0	1	1 m ³ /hr	2020	None	0.25	Unknown
Decarbonator	150,000	1	1,577 m ³ /hr	2007	None	N/A	Survey of High-Recovery and Zero Liquid Discharge Technologies for Water Utilities (2008). Table A2.2.
Dissolved Air Flotation	336,873	0.318	911 m ³ /hr	2007	0.05	0.9999	Unknown
Dual Media Filtration	3,481,852	0.586	30 MGD	2007	0.0507	0.99	Unknown
Electrodialysis Reversal	31,000,000	1	946 m ³ /hr	2016	f(x)	0.9214	Unknown
Energy Recovery	100,000	0.700	4 m ³ /hr	2020	-3.17	N/A	Unknown
Feed Water Tank	2,000,000	0.700	1,736 m ³ /hr	2018	None	N/A	Unknown
Fixed Bed Bioreactor	1,542,008	0.734	1 MGD	2017	0.0309	N/A	Regressed from standard designs for EPA-WBS Biological Treatment with Fixed Bed using pressure vessels from 2017
Injection Well Disposal	22,500,000	0.700	473 m ³ /hr	2011	0.41	N/A	Unknown
Intrusion Mitigation	44,600,000	1	4,750 m ³ /hr	1998	0.0512	N/A	Monterey One Case Study Data
Landfill	1,000,000	0.700	1 m ³ /hr	2020	None	N/A	Unknown
Media Filtration	725,570	0.586	1 MGD	2014	0.00015	0.9999	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Membrane Bioreactor	725,570	0.586	1 MGD	2014	f(Q)	0.9999	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Microfiltration	500,000	1	1 MGD	2014	0.18	0.95	Texas Water Development Board IT3PR Table 3.21
Municipal Drinking	40,299	0.866	1 MGD	2007	f(Q)	N/A	McGivney & Kawamura Figure 5.5.35b
Municipal Wwtp	1,000,000	0.700	100,000 kg/hr	2020	None	N/A	Unknown
Nanofiltration	750,000	1	1 MGD	2014	0.231	0.85	Texas Water Development Board IT3P Table 3.21

Primary Separator	177,516	0.318	5,450 m³/hr	2007	None	0.999	McGivney & Kawamura, Figure 5.5.24c, Cost Estimating Manual for Water Treatment Facilities
Pump	40,299	0.866	1 MGD	2007	0.051	N/A	McGivney & Kawamura Figure 5.5.35b
SMP	1,000,000	0.700	100,000,000 kg/hr	2020	None	N/A	Unknown
Screen	7	1	1 m³/d	2018	None	0.9999	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management. Figure 4.6
Seawater Intake	3,624	0.889	1 m³/hr	2018	f(Q)	N/A	Voutchkov, N. (2018). Desalination Project Cost Estimating and Management, Figure 4.2 + Figure 4.4
Secondary Treatment WWTP	1,000,000	0.700	100,000 kg/hr	2020	0.14	N/A	Unknown
Settling Pond	177,516	0.318	5,450 m³/hr	2007	None	0.9999	McGivney & Kawamura, Figure 5.5.24c + Table 5.2.1, Cost Estimating Manual for Water Treatment Facilities
Sludge Tank	1,000,000	0.700	100,000 kg/hr	2020	None	0.99999	Unknown
Static Mixer	1,698	0.325	1 L/s	2010	None	N/A	Chemical Engineering Design, 2nd Edition. Principles, Practice and Economics of Plant and Process Design. Table 7.2 eq fit to power eq.
Surface Discharge	648	0.876	1 m³/d	2020	f(Q)	N/A	https://www.lenntech.es/processes/surface-water-discharge-of-brine.htm
Tramp Oil Tank	1,000,000	0.700	100,000 kg/hr	2020	None	N/A	Unknown
Tri Media Filtration	725,570	0.586	1 MGD	2014	0.00045	0.9	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Ultrafiltration	500,000	1	1 MGD	2014	0.231	0.95	Texas Water Development Board IT3P Table 3.21
WAIV	110,000	1	64 kg/hr	2020	0.65	0.0001	Unknown
Walnut Shell Filter	725,570	0.586	1 MGD	2014	None	0.99	Texas Water Development Board IT3PR version 2.0, Figure 3.3
Water Pumping Station	24,348	0.867	1 MGD	2007	None	N/A	McGivney & Kawamura Figure 5.5.36b

All parameters are defaults and fully tunable by the user. Source: watertap-org/watertap GitHub repository.